PRAVEEN V 241901082

Exercise 1

Footprinting on the Target Network Using Search Engines, Web Services, and Social Networking Sites

AIM

To gather information about a target organization (domain, IP addresses, DNS records, technologies, emails, and employees) using publicly available sources such as search engines, web services, and social networking sites, without directly interacting with the target systems.

PROCEDURE

Part A: Search Engines (Google Dorking)

1. Open a web browser and go to www.google.com.
2. Choose the target domain (testphp.vulnweb.com).
3. Enter the following search operators one by one:
    - site:testphp.vulnweb.com
    - site:testphp.vulnweb.com filetype:pdf
    - site:testphp.vulnweb.com intitle:“index of”
    - site:testphp.vulnweb.com inurl:login
4. Note the pages, documents and login portals listed.

Part B: Web Services

1. Open who.is or whois.domaintools.com and enter the domain to get registrar, creation date and name server details.
2. Open netcraft.com, use “What’s that site running?” to find the web server, OS and hosting provider.
3. Open shodan.io and search the domain/IP to find open ports and services.
4. Open dnsdumpster.com and enter the domain to get DNS records and subdomains.

5. Open web.archive.org and enter the domain to view archived versions of the site.
6. Open mxtoolbox.com to find mail servers and IP details.

Part C: Social Networking Sites

1. Open LinkedIn and search the company name to find employee names and job titles.
2. Search the organization on Facebook and Twitter/X for official pages and announcements.
3. Check job postings for hints about the technology stack used.
4. In Kali Linux, run theHarvester to collect emails and hostnames: theHarvester -d testphp.vulnweb.com -b google
5. Record all findings in the observation table.

OBSERVATION TABLE

S.No	Tool / Technique	Information Gathered	Observation
1	Google Dork: site:	Indexed pages of the target	Public pages and subdomains found
2	Google Dork: filetype:pdf	Public documents	Documents with internal details found
3	Whois Lookup	Registrar, creation date, name servers	Domain ownership details obtained
4	Netcraft	Web server, OS, hosting provider, IP	Server technology identified
5	Shodan	Open ports, services, banners	Exposed services listed
6	DNSdumpster	DNS records, subdomains, MX records	Subdomains and mail servers identified
7	Wayback Machine	Archived versions of website	Old pages and structure viewed
8	LinkedIn	Employee names, designations	Key personnel identified
9	Facebook / Twitter	Official pages, posts	Organizational activity

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S.No	Tool / Technique	Information Gathered	Observation
			found
10	theHarvester	Emails, subdomains, hostnames	Email addresses and hosts collected

OUTPUT

Google search with site: and filetype: operators listed the indexed pages and documents of the target domain.

Whois lookup displayed the registrar, registration date and name server details.

Netcraft report showed the web server, operating system and hosting provider.

Shodan displayed open ports, services and banners of the target IP.

DNSdumpster listed DNS records, subdomains and mail servers.

Wayback Machine displayed archived snapshots of the website.

LinkedIn and social media searches revealed employee names, designations and official pages.

theHarvester output:

[*] Target: testphp.vulnweb.com

[*] Emails found: 2

admin@testphp.vulnweb.com

info@testphp.vulnweb.com

[*] Hosts found: 3

www.testphp.vulnweb.com

mail.testphp.vulnweb.com

ftp.testphp.vulnweb.com

RESULT

Thus, footprinting of the target network was successfully performed using search engines, web services and social networking sites. Domain

registration details, IP addresses, DNS records, subdomains, web server technology, email addresses and employee details were collected using passive reconnaissance techniques.